

Math 407 Section 1

Fall 2026

Exam I

February 11, 2026

Time Limit: 1 Hour 50 Minutes

Name (Print): _____

Student ID: _____

This exam contains 8 pages (including this cover page) and 7 problems. Check to see if any pages are missing. Enter all requested information on the top of this page, and put your initials on the top of every page, in case the pages become separated.

You may *not* use your books or notes on this exam.

You are required to show your work on each problem on this exam. The following rules apply:

- **Organize your work**, in a reasonably neat and coherent way, in the space provided. Work scattered all over the page without a clear ordering will receive very little credit.
- **Mysterious or unsupported answers will not receive full credit.** A correct answer, unsupported by calculations, explanation, or algebraic work will receive no credit; an incorrect answer supported by substantially correct calculations and explanations might still receive partial credit. This especially applies to limit calculations.
- If you need more space, use the back of the pages; clearly indicate when you have done this.
- If the problem asks for a proof, be sure to carefully justify your work, including any theorems from class.

Note: you may NOT use a theorem or result from class to prove something when it makes the problem entirely trivial. If you are unsure whether a particular theorem or result is allowed, just ask!

Problem	Points	Score
1	10	
2	10	
3	10	
4	10	
5	10	
6	10	
7	10	
Total:	70	

Do not write in the table to the right.

1. (10 points) **TRUE or FALSE!** Write TRUE if the statement is true. Otherwise, write FALSE. Your response should be in ALL CAPS. No justification is required.

(a) The group $\mathbb{Q}$ with binary operation $+$ is a cyclic group

(b) There are exactly two primitive 4'th roots of unity

(c) A cyclic group must be Abelian

(d) $xy + xz + yz$ is an elementary symmetric polynomial in three variables

(e) The order of the permutation $(1\ 3\ 4\ 5\ 2)$ is 5

2. (10 points)

(a) Write the definition of a polynomial $p(z_1, \dots, z_n)$ being symmetric

(b) State Vieta's Theorem

(c) Suppose that z_1, z_2, z_3, z_4 are the solutions of

$$z^4 + 3z^3 + 2z^2 + 5z + 7 = 0.$$

Determine the value of

$$z_1^2 + z_2^2 + z_3^2 + z_4^2$$

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3. (10 points) Let G be a group and suppose that $\emptyset \neq H \subseteq G$ satisfies $ab^{-1} \in H$ for all $a, b \in H$. Prove that H is a subgroup of G .

4. (10 points) Consider the set

$$G = (\mathbb{Z}_7)^\times = \{x \in \mathbb{Z}_7 : x \neq 0\}.$$

with the binary operation given by multiplication modulo 7.

- (a) Show that G has an identity element

- (b) For each element of G , determine the inverse.

- (c) Prove that G is a cyclic group

- (d) Explain why G is isomorphic to $\mathbb{Z}_6$

5. (10 points)

(a) Write down the definition of an n 'th root of unity and a primitive n 'th root of unity.

(b) Write down all 15'th roots of unity. How many are there?

(c) Which of the 15'th roots of unity are primitive? How many are there?

6. (10 points) Consider the group $G = S_9$.

(a) Rewrite the following permutation in cycle notation

$$\begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 & 9 \\ 8 & 5 & 7 & 6 & 3 & 2 & 1 & 9 & 4 \end{pmatrix}$$

(b) Rewrite the following permutation in standard notation

$$(2\ 4\ 6)(1\ 3)(5\ 7\ 9)$$

(c) How many elements are in the cyclic subgroup $\langle \sigma \rangle$ generated by the permutation

$$\sigma = (1\ 2\ 3)(4\ 5)$$

7. (10 points)

(a) Write down the definition of the sign of a permutation in S_n .

(b) Prove that the following set is a subgroup of S_n

$$A_n = \{\sigma \in S_n : \sigma \text{ is even}\}$$

(c) Prove that for all integers $n > 1$,

$$|A_n| = \frac{n!}{2}$$